

Test 23-24_MAS Exam December



Test ID: 206636 (In progress)

Folder: /Top/SIS/XB_0024

Version: 4.1

Randomised: No

Last modified: Tuesday, 2 december 2025 16:44:00

Number of questions: 29

Blocks: Fixed

Display questions once: No

Tools: Digital scratch paper

Test time: -

Cut-off: Split linear: 60.00 % (Exclude chance score)

Maximum score: 100 pt.

Chance score: 3.82 pt. / 4%

Mark scale: One decimal

Grading series: 1 - 10

Scoring rule: No point deduction for incorrect answer

In test set with: -

Test instruction

Welcome to the exam of the course Multi-Agent Systems (XB_0024). This exam will test your knowledge and understanding of the material provided to you and presented in the lectures, the book of Blackburn, Bos and Striegnitz Learn Prolog Now! (Chapters 1 to 6, 10 and 11), and the MARBEL Programming Guide. It is not allowed to use materials such as books, scientific papers or slides during the exam. You will have 2:45 hours to complete the exam.

The following tools are permitted:

- Pen and drafting paper.

For questions about the content of the test, you can contact the teacher. Information about the review of the exam is provided via Canvas.

If you have **not** signed up for this exam, you will not receive a result. Through VUnet you can object to the fact that you can no longer sign up after the expiry of the registration deadline (and the fact that you will not receive a result for this exam). Submit your appeal online within one week after the exam. More information can be found at www.vu.nl/intekenen.

Good luck!

Welkom bij de toets van het vak Multi-Agent Systems (XB_0024). Deze toets test je kennis en begrip van de stof als gepresenteerd in de colleges, het boek van Blackburn, Bos en Striegnitz Learn Prolog Now! (Hoofdstukken 1 tot 6, 10 en 11), en de MARBEL Programming Guide. Het is niet toegestaan om hulpmiddelen zoals boeken, publicaties en slides te gebruiken tijdens de toets. Je hebt 2:45 uur om de toets te maken.

Bij deze toets mag je gebruik maken van de volgende hulpmiddelen:

- Pen en kladpapier.

Voor vragen over de inhoud van de toets kun je bij de docent terecht. Informatie over de toetsinzage wordt via Canvas verstrekt.

Als je **niet** hebt ingetekend voor dit tentamen, ontvang je geen resultaat. Via VUnet kun je bezwaar maken tegen het feit dat je niet meer kunt intekenen na het verstrijken van de intekendeadline (en je dus geen resultaat krijgt voor dit tentamen). Dien je bezwaarschrift online in binnen één week na het tentamen. Meer informatie vind je op www.vu.nl/intekenen.

Veel succes!

Prolog queries

Question order: Random

The following questions each give a Prolog query, after which you are asked to fill in the outcome that Prolog will give based on the query.

Question 1 – 23-24_prolog_query_1 – 402752.2.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 14:56

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

?- Life = life.

Life = life.

Life = life

Life=life.

Life=life

Life= life.

Life= life

Life =life.

Life =life

Question 2 – 23-24_prolog_query_2 – 402747.3.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 14:59

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

?- h(X,y) = h(X,z).

false.

false

Question 3 – 23-24_prolog_query_3 – 402748.3.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 15:04

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

?- f(g(X,y,Z)) = f(g(Z,Y,x)).

X = Z, Z = x, Y = y.

X = Z, Z = x, Y = y

X=Z, Z=x, Y=y.

X=Z, Z=x, Y=y

X=Z,Z=x,Y=y.

X=Z,Z=x,Y=y

X=Z, Z = x, Y = y.

X=Z, Z = x, Y = y

X=Z, Z=x, Y = y.

X=Z, Z=x, Y = y

X=Z, Z = x, Y=y.

X=Z, Z = x, Y=y

Question 4 – 23-24_prolog_query_4 – 402751.2.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 15:05

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

?- 4 * 4 is 16.

false.

false

Question 5 – 23-24_prolog_query_5 – 402742.2.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 15:06

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

?- X is 8 / 2.

X = 4.

X = 4

X=4.

X=4

X= 4.

X= 4

X =4.

X =4

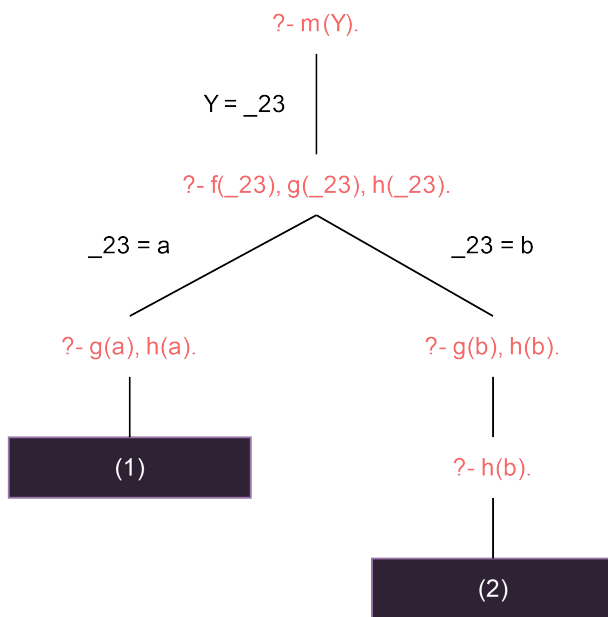
Prolog search

Question order: Random

The questions in this block will be based on the following knowledge base:

f(a).
f(b).
g(b).
h(a).
h(X) :- k(X), l(X).
k(a).
k(b).
l(a).
l(b).
m(X) :- f(X), g(X), h(X).

Part of the search tree for the query `?- m(Y)` is given:



Tip: keep the introduction open when answering the following questions, by clicking the 'show block intro' button above the question.

Question 6 – 23-24_prolog_search_1 – 402808.1.0

Question type: Multiple choice

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.40 pt. / 20%

Status:

Last modified: 11/12/2023 16:33

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

The search tree depicted above is incomplete. If it was complete, how many branches would there be?

- ☒ **A** 2
- ☐ **B** 3
- ☐ **C** 4
- ☐ **D** 5
- ☐ **E** Infinite

Question 7 – 23-24_prolog_search_2 – 402809.2.0

Question type: Multiple choice

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Random

Partial scoring: No

Maximum score: 2

Chance score: 0.25 pt. / 13%

Status:

Last modified: 11/12/2023 16:36

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Of the two branches depicted in the search tree, what are their outcomes?

- ☐ **A** left: success, right: success
- ☐ **B** left: failure, right: failure
- ☒ **C** left: failure, right: success
- ☐ **D** left: success, right: failure
- ☐ **E** left: success, right: infinite
- ☐ **F** left: infinite, right: success
- ☐ **G** left: failure, right: infinite
- ☐ **H** left: infinite, right: failure

Question 8 – 23-24_prolog_search_3 – 402810.1.1

Question type: Multiple choice

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 2

Chance score: 0.50 pt. / 25%

Status:

Last modified: 11/12/2023 16:40

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Which strategy is being applied to move from the starting query $?- m(Y)$ to the following query $?- f(_{23}), g(_{23}), h(_{23})$?

- ☒ **A** Backward chaining
- ☐ **B** Linear traversal
- ☐ **C** Backtracking
- ☐ **D** Tree expansion

Question 9 – 23-24_prolog_search_4 – 402813.1.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 5

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 16:44

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Fill in box (1) in the search tree displayed above.

failure

false

failure.

false.

Question 10 – 23-24_prolog_search_5 – 402815.2.0

Question type: Fill in

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: No

Maximum score: 5

Chance score: 0.00 pt. / 0%

Status:

Last modified: 11/12/2023 16:47

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Fill in box (2) in the search tree displayed above.

?- k(b), l(b).

?- k(b), l(b)

k(b), l(b).

k(b), l(b)

?- k(b),l(b).

?- k(b),l(b)

k(b),l(b).

k(b),l(b)

?-k(b), l(b).

?-k(b), l(b)

?-k(b),l(b).

?-k(b),l(b)

Prolog concepts

Question order: Fixed

The following questions address your knowledge of Prolog concepts. Select none, one, or more answers from a list of answers options.

Question 11 – 23-24_prolog_concepts_1 – 402829.2.0

Question type: Multiple response

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Random

Partial scoring: Yes

Maximum score: 2

Chance score: 0.67 pt. / 33%

Status:

Last modified: 11/12/2023 17:19

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

When is a predicate `predQ` reachable from a predicate `predR` in a program `P`? Select all the correct options.

- ☐ **A** There is a rule in `P` the head of which uses `predQ` and the body of which contains `predR`
- ☒ **B** There is a rule in `P` the head of which uses `predR` and the body of which contains `predQ`
- ☒ **C** There is a predicate `predW` such that `predW` is reachable from `predR` in `P`, and `predQ` is reachable from `predW` in `P`.
- ☐ **D** There is a predicate `predW` such that `predW` is reachable from `predQ` in `P`, and `predR` is reachable from `predW` in `P`.

Question 12 – 23-24_prolog_concepts_2 – 402834.3.0

Question type: Multiple response

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Random

Partial scoring: Yes

Maximum score: 2

Chance score: 0.67 pt. / 33%

Status:

Last modified: 11/12/2023 17:33

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

What makes the use of an accumulator more efficient during list recursion than a simple recursive procedure? Select all the relevant correct statements.

- ☐ **A** Tail-recursion makes simple recursion less efficient. An accumulator circumvents this.
- ☒ **B** Tail-recursion makes an accumulator more efficient.
- ☒ **C** The core of the inefficiency of a simple recursive procedure lies in having to access all the created stacks again when a solution is found. This is a waste of computational resources.
- ☐ **D** The core of the efficiency gain of an accumulator is its ability to skip irrelevant queries and thus it reaches a solution faster. This saves computational resources.

Programming in Prolog

Question order: Random

The following two questions will be programming exercises. You will have to create a Prolog program, using correct Prolog syntax, defining one or more predicates to answer each question.

Question 13 – 23-24_prolog_programming_1 – 402846.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 10

Chance score: 0.00 pt. / 0%

Status:

Last modified: 18/12/2023 10:04

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 500

Word count: No

Create a predicate `lastN/3` that is true if `List2` contains the last `N` elements of `List1`. For example, `lastN(List1, N, List2)` is true if `List1 = [1,2,3,4,5]`, `N = 3` and `List2 = [3, 4, 5]`.

The only swi-prolog built-in predicate you are allowed to use is `length/2`.

`length(?List, ?Length)`: *true if Length represents the number of elements in List.*

Grading instruction

Answer:

`lastN(L,N,R) :- length(L,X), X1 is X-N, lastT(L,X1,R).`

`lastT(L,0,L).`

`lastT([_|T],X,L) :- X>0, X2 is X-1, lastT(T,X2,L).`

10 points

Criterion 1 (Number of points: 1)

Head consists of 2 lists and a size indicator. For example: `lastN(L,N,R)`

Criterion 2 (Number of points: 1)

Solution contains recursion

Criterion 3 (Number of points: 2)

Base clause stops recursion at the appropriate moment. For example: `lastT(L,0,L).`

Criterion 4 (Number of points: 3)

Recursive clause recursively takes last element from list (if there still needs to be taken a step). For example: `lastT([_|T],X,L) :- X>0, X2 is X-1, lastT(T,X2,L).`

-1 pnt without `X>0` check.

Criterion 5 (Number of points: 3)

Step to calculate the number of required recursive steps and use a helper function. For example: `lastN(L,N,R) :- length(L,X), X1 is X-N, lastT(L,X1,R).`

Question 14 – 23-24_prolog_programming_2 – 402849.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 10

Chance score: 0.00 pt. / 0%

Status:

Last modified: 18/12/2023 10:07

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 500

Word count: No

Create a predicate `remdup/2` that is true if `list2` is a subset of `list1` without any duplicates. For example, `remdup(List1, List2)` is true if `List1 = [a,b,b,c,d,d,b,e]` and `List2 = [a,b,c,d,e]`. Use an accumulator in your solution.

You are only allowed to use the built-in swi-prolog predicate `member/2`.

`member(?Elem, ?List): true if Elem is a member of List.`

Grading instruction

Answer:

`remdup(In, Out) :- dupacc(In, [], Out).`

`dupacc([], A, A).`

`dupacc([H|T], A, L) :- member(H, A), dupacc(T, A, L).`

`dupacc([H|T], A, L) :- dupacc(T, [H|A], L).`

10 points

Criterion 1 (Number of points: 2)

Use of accumulator with arity 3. For example `dupacc(List1, Accumulator, List2)`.

Criterion 2 (Number of points: 2)

Call accumulator from main wrapper predicate.

Criterion 3 (Number of points: 2)

Recursively step through the elements of a list and transfer the accumulator list to the output list when the input list is empty. For example, `dupacc([], A, A)`.

Criterion 4 (Number of points: 4)

Solution includes differentiating between membership and non-membership cases to include or exclude an element from adding it to the accumulator list. For example:

`dupacc([H|T], A, L) :- member(H, A), dupacc(T, A, L).`

`dupacc([H|T], A, L) :- not(member(H,A)), dupacc(T, [H|A], L).`

-1 pnt if two rules are not mutually exclusive.

Agents and MARBEL: Basics

Question order: Fixed

This part tests your basic knowledge of agents and MARBEL programming concepts.

Question 15 – 23-24_MARBEL_Basics_1a1 – 402157.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:02

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 20

Word count: Yes

Environments that agents can connect to can be classified according to different properties.

Explain what it means for an environment to be a *single/multi-agent* environment.

Grading instruction

Criterion 1 (Number of points: 1)

There is (more than) one agent...

Criterion 2 (Number of points: 1)

...performing actions changing the environment.

Question 16 – 23-24_MARBEL_Basics_1a2 – 402158.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:02

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 15

Word count: Yes

Apart from the single/multi-agent property, mention two further properties that environments can have which are relevant for designing agents.

Grading instruction

Criterion 1 (Number of points: 1)

One of:

- fully/partially observable
- static/dynamic
- discrete/continuous
- deterministic/stochastic

Criterion 2 (Number of points: 1)

Another one of:

- fully/partially observable
- static/dynamic
- discrete/continuous
- deterministic/stochastic

Question 17 – 23-24_MARBEL_Basics_1b – 402159.1.2

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 4

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:03

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 30

Word count: Yes

In a single agent environment, that agent may have full control over what happens in an environment and therefore the agent can always be sure what the effects of performing an action in that environment are.

Mention two (other) specific environment properties that an environment must have for an agent to have full control.

Grading instruction

Criterion 1 (Number of points: 2)

One of:

- The environment is static (changes cannot happen because of the environment)
- The environment is deterministic (an agent knows the outcome of the effects of actions).

Both of these properties are needed. Note that an environment does not need to be fully observable for an agent to have full control; that answer is wrong.

Criterion 2 (Number of points: 2)

Another one of:

- The environment is static (changes cannot happen because of the environment)
- The environment is deterministic (an agent knows the outcome of the effects of actions).

Both of these properties are needed. Note that an environment does not need to be fully observable for an agent to have full control; that answer is wrong.

Question 18 – 23-24_MARBEL_Basics_1c1 – 402165.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:03

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 30

Word count: Yes

What is a so-called *send once* percept?

Grading instruction

Criterion 1 (Number of points: 1)

Send once percepts are percepts that are sent only once...

Criterion 2 (Number of points: 1)

... when the agent first connects to an environment.

Question 19 – 23-24_MARBEL_Basics_1c2 – 402166.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:04

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 30

Word count: Yes

What problem would the agent have when the *replace* channel handler would be used to process send once percepts?

Grading instruction

Criterion 1 (Number of points: 2)

The percepts would be removed again from the agent's database (which might make adding the percept information initially useless, as the information will never be sent again).

Question 20 – 23-24_MARBEL_Basics_1c3 – 402177.1.0

Question type: Multiple choice

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Random

Partial scoring: No

Maximum score: 2

Chance score: 0.67 pt. / 33%

Status:

Last modified: 10/12/2023 20:29

Attributes: *Taxonomie/Taxonomy* 01 Kennis/Knowledge

Which channel handler can be best used to handle send once percepts?

- A** The *add* handler.
- B** The *replace* handler.
- C** The *update* handler.

Question 21 – 23-24_MARBEL_Basics_1c4 – 402186.1.0

Question type: Multiple choice

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Random

Partial scoring: No

Maximum score: 4

Chance score: 0.67 pt. / 17%

Status:

Last modified: 10/12/2023 20:42

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Where should code for handling a *send once* percept be placed (in what type of file, and in which part of that file)?

- A** In a *mas2g* file, as part of the dynamic predicate declaration.
- B** In a Prolog file, as part of the dynamic predicate declaration.
- C** In a *mod2g* file, as part of the use clauses.
- D** In a *mod2g* file, as part of the program section.
- E** In a Prolog file, as part of the launch policy section.
- F** In a *mas2g* file, as part of the agent definition.

Programming in MARBEL

Question order: Fixed

This assignment is about a (toy) vacuum cleaning robot that is deployed in an office space with 5 offices. The robot is able to collect dust from a single office space and then has to empty its bin before it can clean another office. The robot can empty its bin in the public area of the office space and move from there to any of the 5 offices. You may assume that the robot can always move freely from one place to another in the office space. Carefully analyze the MARBEL agent system in Figure 1 below.

Also make sure to follow the instructions given in Remark 1 in Figure 1.

Remark 1: Important for the MARBEL programming questions: When you add code in the program templates, clearly specify the position in the code where you want to add something.
If you want to *modify* a line, for example, number 11, in a module named 'file.mod2g', and replace it with code '...', use the following notation:
file.mod2g 11 ...
If you want to place *new* code '...' in some module file between two line numbers, for example, 23 and 24, use the following notation:
file.mod2g 23.1 ...
file.mod2g 23.2 ...
file.mod2g 23.3 ...

```
----- dustRobot.mas2g -----
1 use "sensors-and-actuators.jar".
2
3 define dustRobot {
4   use dustInit for init.
5   use dustAction for decisions.
6
7   ...
8   ...
9 }
10
11 launchpolicy {
12   when * launch dustRobot.
13 }
----- dust.pl -----
1 :- dynamic at/1, binEmpty/o, office/1.
----- dustInit.mod2g -----
1 use dust.
2
3 module dustInit {
4   ...
5
6   if true then insert(binEmpty).
7 }
----- dustAction.mod2g -----
1 use dust.
2
3 module dustAction {
4   % If the bin is empty...
5   if binEmpty then {
6     % and if there is dust in an office...
7     if dust(Office) then {
8       % and if at the right place, then clean office.
9       if ... then clean + delete(binEmpty).
10      % or else go to office for cleaning.
11      if true then move(Office).
12    }
13    % no dust anywhere, let's move to public area.
14    if true then move(public).
15  }
16  % else empty bin when in public area.
17  if at(public) then empty.
18  % or else, last but not least, go back to public area.
19  if true then move(public).
20 }
```

Figure 1: The dustRobot agent.

Tip: keep the introduction open when answering the following questions, by clicking the 'show block intro' button above the question.

Question 22 – 23-24_MARBEL_Programming_2a – 402203.4.0

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 21/12/2023 15:17

Attributes: *Taxonomie/Taxonomy* 03 Toepassen/Apply

Word count: 50

Word count: Yes

The public office space area is simply labelled as the public area. Add an action rule to the code in the most appropriate place (indicate in which file) to make sure that the agent knows this public office space area exists, i.e. `office(public)` is in its database.

Grading instruction

Criterion 1 (Number of points: 1)

dustInit.mod2g 4

Criterion 2 (Number of points: 1)

```
if true then insert(office(public)).
```

It is also fine to use `forall-do` instead of `if-then`. If some mistake(s) in rule but more or less still OK, deduct 1pt. 1pt for putting rules in correct place, i.e. the init module. 2pt for each rule.

Question 23 – 23-24_MARBEL_Programming_2aa – 405935.2.0

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 3

Chance score: 0.00 pt. / 0%

Status:

Last modified: 21/12/2023 15:11

Attributes: *Taxonomie/Taxonomy* 03 Toepassen/Apply

Word count: 50

Word count: Yes

Besides the public office space there are five other office spaces which are known simply by their numbers 1 to 5. Add an action rule to the code in the most appropriate place (indicate in which file) for adding all these other office areas to the agent's database using the `office/1` predicate.

Grading instruction

Criterion 1 (Number of points: 1)

dustInit.mod2g 4

Criterion 3 (Number of points: 2)

```
dustInit.mod2g 4.1 if true then insert(office(1), office(2), office(3), office(4), office(5)).
```

It is also fine to use `forall-do` instead of `if-then`. If some mistake(s) in rule but more or less still OK, deduct 1pt. 1pt for putting rules in correct place, i.e. the init module. 2pt for each rule.

Question 24 – 23-24_MARBEL_Programming_2b – 402215.1.2

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 4

Chance score: 0.00 pt. / 0%

Status:

Last modified: 21/12/2023 15:15

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 50

Word count: Yes

Assume that `dust/1` is a *send always* percept informing the agent that when it receives, for example, `dust(2)` that office 2 has dust. Add a channel for handling this percept with the appropriate handler to the code, and indicate where you would add this. What else needs to be added to the program to make the inclusion of `dust/1` work without problems (again, also indicate where)?

Grading instruction

Criterion 1 (Number of points: 2)

`dustRobot.mas2g`, line 7 `replace dust/1`.

Criterion 2 (Number of points: 2)

The predicate needs to be declared. In `dust.pl` line 1 `:- dynamic at/1, binEmpty/0, dust/1, office/1`.

Correct place `mas2g` (1p) and correct handler 'replace' (1p).

Indication that `dust/1` predicate needs to be declared (1p) and this needs to be added to line 1 in `dust.pl` (1p).

Just the phrase: `dust.pl` line 1 `:- dynamic at/1, binEmpty/0, dust/1, office/1`. is also OK (=2p).

Question 25 – 23-24_MARBEL_Programming_2c – 402220.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 2

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:36

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 50

Word count: Yes

The `at/1` predicate is a *send on change* percept informing the agent about whether it is at a particular office area. Add an appropriate handler for this percept and indicate where this should be put in the code.

Grading instruction

Criterion 1 (Number of points: 1)

`dustRobot.mas2g` line 8

Criterion 2 (Number of points: 1)

`update at/1`.

Question 26 – 23-24_MARBEL_Programming_2d – 402222.1.1

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 3

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:35

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 50

Word count: Yes

Complete the code in `dustAction.mod2g` line 9 and replace the `. . .` by specifying a query. Do not introduce any new predicates but use one or more of the predicates listed in the code listing (see Figure 1) to specify the query. Note that you just need to specify the query!

Grading instruction

Criterion 1 (Number of points: 3)

Replace `...` with `at(Office)`.

Any issue deduct one point.

Question 27 – 23-24_MARBEL_Programming_2e – 402226.1.0

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 4

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:26

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 50

Word count: Yes

The status of the robot's bin is not received via a percept from the environment and the agent needs to keep track of that status itself. It already removes the fact the bin is empty after cleaning an office that has dust (check the code listing in Figure 1).

The agent does not (re)insert it after the agent has emptied the bin. Fix this problem. Indicate which code you changed how and where.

Grading instruction

Criterion 1 (Number of points: 4)

In `dustAction.mod2g` line 17 `if at(public) then empty + insert(binEmpty)`.

1pt for correct location.

1pt for using the `+` operator.

2pts for the insert action.

Deduct 1pt if a (minor) mistake is made.

Question 28 – 23-24_MARBEL_Programming_2f – 402227.1.0

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 6

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:28

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 500

Word count: No

Why do we not have to add `not (binEmpty)` to the query of the rule in line 17 in `dustAction.mod2g`? Explain this by reasoning about the other rules that are in this module.

Grading instruction

Criterion 1 (Number of points: 2)

The rule at line 5 already checks for `binEmpty`.

Criterion 2 (Number of points: 2)

Moreover, if the bin is empty, i.e. `binEmpty` holds, the rule at line 17 will never be reached...

Criterion 3 (Number of points: 2)

... because in that case line 14 will always be applicable.

Question 29 – 23-24_MARBEL_Programming_2g – 402230.1.0

Question type: Open-ended

Pre-test item: No

Folder: /Top/SIS/XB_0024

Folder description: Multi-Agent Systems

Answer option order: Fixed

Partial scoring: Yes

Maximum score: 8

Chance score: 0.00 pt. / 0%

Status:

Last modified: 10/12/2023 21:33

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Word count: 500

Word count: No

Suppose that the robot can always perform the `empty` action successfully, even when its bin is already empty. If we remove the action rule on line 6 in the `dustInit.mod2g` file, assuming that you fixed the bin status problem above, would the `dustRobot` agent still be able to function properly? Carefully explain your answer by providing all the relevant details about why the robot will still (or will not) be able to clean offices.

Grading instruction

Criterion 1 (Number of points: 1)

Yes

Criterion 2 (Number of points: 2)

First, note that initially the nested rule in line 5 (ending at line 15) will never be entered.

Criterion 3 (Number of points: 1)

Moreover, note that the robot will be either at the public space area (case 1) or not (case 2).

Criterion 4 (Number of points: 1)

In the former case 1, it will execute the `empty` action (without any problems) and insert that the bin is empty in its database.

Criterion 5 (Number of points: 1)

In the latter case 2, line 19 will be executed and the former case 1 applies again.

Criterion 6 (Number of points: 1)

In any case, the fact that the bin is empty will eventually be added to the agent's database...

Criterion 7 (Number of points: 1)

... and the robot will be able to clean offices by means of the nested rule on line 5.

Test matrix

No matrix defined